

# FPGA-based Adaptive Texture-Aware Stereo Matching Accelerator for Co-axial RGB–NIR Sensors

Chenghe Zhang  
Tiangong University  
Tianjin, China

2431101264@tiangong.edu.cn

Qingzeng Song  
Tiangong University  
Tianjin, China

qingzengsong@163.com

Yongjiang Xue  
Tiangong University  
Tianjin, China

xueyongjiang@tiangong.edu.cn

Weigang Kong  
Tiangong University  
Tianjin, China  
kwg19970502@gmail.com

Fei Qiao  
Tsinghua University  
Beijing, China  
qiaofei@tsinghua.edu.cn

## Abstract

Edge robots require reliable all-day depth perception, but deploying RGB–NIR heterogeneous sensor fusion is severely constrained by strict size, weight, and power limits. Furthermore, traditional cross-modal registration and dynamic vertical residual compensation rely on external memory buffering, which stalls the continuous hardware dataflow. This paper presents a hardware–algorithm co-designed, DDR-less stereo matching accelerator tailored for co-axial RGB–NIR sensors to enable robust all-day edge vision. By employing a fully pipelined datapath with on-chip line buffers, the design bypasses all external memory dependencies. A cost-domain adaptive texture fusion (CATF) module reuses the Census transform register array to extract texture confidence, adaptively modulating semi-global matching penalty parameters and cross-modal fusion weights with near-zero overhead. To ensure robustness against mechanical vibrations and thermal drifts, an epipolar redundancy mechanism dynamically searches adjacent scanlines, tolerating up to 1-pixel vertical residuals. Implemented on a Zynq-7020 SoC, the system processes 720×540 video streams at 128 FPS. Consuming 35,868 LUTs and 2.43 Mb BRAM, it reduces the mean absolute error by up to 19.9% in all-day scenarios compared to OpenCV SGBM baselines, achieving an effective balance between robust multimodal perception and hardware efficiency.

## CCS Concepts

• **Hardware** → **Hardware accelerators**; • **Computer systems organization** → **Special purpose systems**.

## Keywords

FPGA, Stereo Matching, Multi-modal Stereo, Hardware–Algorithm Co-design, All-Day Edge Vision

## ACM Reference Format:

Chenghe Zhang, Qingzeng Song, Yongjiang Xue, Weigang Kong, and Fei Qiao. 2026. FPGA-based Adaptive Texture-Aware Stereo Matching Accelerator for Co-axial RGB–NIR Sensors. In *Great Lakes Symposium on VLSI 2026*



This work is licensed under a Creative Commons Attribution 4.0 International License. *GLSVLSI '26, Canandaigua, NY, USA*  
© 2026 Copyright held by the owner/author(s).  
ACM ISBN 979-8-4007-2431-2/26/06  
<https://doi.org/10.1145/3787109.3815247>

(*GLSVLSI '26*), June 22–24, 2026, Canandaigua, NY, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3787109.3815247>

## 1 Introduction

All-day depth perception is essential for mobile robots navigating unstructured edge environments. Unlike long-range autonomous driving, edge robots operate in complex proximity scenarios (e.g., indoor corridors, tunnel exits) featuring illumination variations and textureless surfaces. Under these extreme conditions, traditional RGB-only stereo vision degrades significantly. Fusing RGB and near-infrared (NIR) modalities effectively compensates for these blind spots. However, deploying robust multimodal fusion algorithms on edge robots is strictly constrained by Size, Weight, and Power (SWaP) budgets. Consequently, Field-Programmable Gate Arrays (FPGAs) have emerged as the compelling hardware choice due to their superior energy efficiency and microsecond-level deterministic latency.

However, deploying high-accuracy multimodal stereo matching at the edge faces three practical challenges. First, the cyclic dependency between cross-modal registration and depth estimation prohibits real-time dynamic alignment [12]. Second, mechanical and thermal drifts inevitably introduce dynamic vertical residuals despite static calibration [9, 19]. Traditional DDR-based frame buffering for image remapping disrupts the FPGA's full-pipeline dataflow [11]. Finally, simple pixel-level fusion suffers from high-frequency cancellation, [20], while network-based fusion exhausts edge logic resources [18].

This paper proposes an FPGA-based adaptive texture-aware stereo matching architecture, optimized specifically for co-axial RGB–NIR sensors. The main contributions are as follows:

- (1) Leveraging the physical pixel-alignment of co-axial sensors, we propose a DDR-less, line-buffer-based streaming architecture that achieves real-time multimodal disparity computation for heterogeneous data.
- (2) We propose a hardware-friendly Cost-domain Adaptive Texture Fusion (CATF) framework. By reusing the Census transform register array to extract local texture confidence, it adaptively modulates fusion weights and SGM  $P_2$  penalty parameters, effectively improving disparity continuity in textureless and illumination-variant regions.
- (3) Targeting dynamic vertical residuals caused by mechanical vibrations and thermal drifts, we design a configurable Epipolar Redundancy Mechanism (ERM) within the cost calculation stage. By searching adjacent scanlines to compensate for vertical offsets up

to 1 pixel, it ensures system robustness under non-ideal alignment conditions.

Described in Verilog HDL with register retiming optimizations, our architecture is implemented on a Zynq-7020 SoC. It achieves real-time  $720 \times 540$  processing at 128 FPS (64-pixel disparity), consuming only 35,868 LUTs, 2.43 Mb BRAM, and 0.825 W. Evaluated on the CVPR 2025 Pixel-aligned RGB-NIR Stereo Dataset [12], it reduces MAE by 16.0% (daytime) and 19.9% (nighttime) compared to OpenCV SGBM, striking an effective balance between robust multimodal perception and hardware efficiency.

## 2 Related Work

### 2.1 Stereo Vision Hardware Architectures

While deep learning approaches have recently achieved dominant accuracy in stereo matching [15, 22], their high resource demands preclude real-time deployment on power-constrained edge platforms ( $< 5$  W). Consequently, Semi-Global Matching (SGM) [7] combined with robust cost metrics (e.g., Census or SAD-Census) [5, 11, 23, 24] remains the de facto standard for real-time, low-power FPGA implementations.

The primary bottleneck in FPGA-based SGM lies in the memory-intensive cost aggregation and frame buffering. Early architectures [1] achieved real-time processing but incurred prohibitive logic and external DDR overheads. Recent optimizations, such as time-sharing [17] or region-based pipelines [4], achieve high frame rates with minimal logic. However, these rigid architectures struggle with complex demands like multimodal fusion or dynamic calibration shifts unless they revert to external DDR.

### 2.2 Adaptive Constraints and Epipolar Residuals

In the SGM algorithm, globally fixed penalty parameters ( $P_1, P_2$ ) struggle to balance the smoothing requirements of textureless regions and object boundaries. Although edge-based penalties [2, 8] improve continuity, their dedicated detection modules inflate logic and memory overhead. Moreover, they often mistake shadows or highlights for physical boundaries, causing depth artifacts in complex lighting.

Moreover, thermal and mechanical drifts in physical deployments introduce dynamic epipolar residuals [9]. Although expanding the vertical search space during cost calculation [19] can theoretically tolerate imperfect rectification, indiscriminate global searching inflates mismatch rates. Consequently, a resource-efficient, texture-aware mechanism that simultaneously handles adaptive penalization and dynamic residuals is critically required.

### 2.3 RGB-NIR Multimodal Fusion

RGB sensors suffer severe Signal-to-Noise Ratio (SNR) degradation in low light, while NIR struggles on textureless surfaces (e.g., white walls). To achieve all-day perception, RGB-NIR fusion is essential. However, existing multimodal systems typically employ discrete camera layouts [10, 21], introducing a “chicken-and-egg” alignment paradox. Resolving this requires complex online registration [3, 13] or DDR-heavy offline remapping [11, 14], both disrupting FPGA streaming and inflating logic overhead.

While emerging prism-based co-axial sensors [12] eliminate geometric registration barriers, efficiently fusing data with massive spectral discrepancies remains challenging. Conventional multi-scale or filtering fusion [6, 16] either demands excessive memory or introduces halo artifacts at gradient reversals (where RGB and NIR polarities invert). Conversely, CNN- or GAN-based fusion networks [18] exceed the power and bandwidth budgets of edge devices. Thus, a lightweight, memory-efficient adaptive fusion strategy tailored for co-axial streaming architectures is urgently needed.

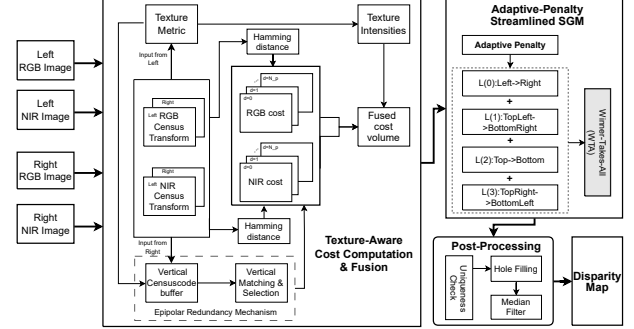


Figure 1: Overall architecture of the proposed hardware-aware RGB-NIR stereo matching algorithm.

## 3 Hardware-Aware Adaptive Texture Fusion Architecture

This section presents an FPGA-based streaming architecture tailored for co-axial RGB-NIR stereo sensors. Because incident light is physically split via a dichroic prism, the heterogeneous inputs are inherently pixel-aligned. This completely bypasses complex cross-modal spatial registration, circumventing traditional memory-wall bottlenecks.

To address the low SNR and dynamic epipolar shifts in unconstrained edge environments, we propose the **Cost-domain Adaptive Texture Fusion (CATF)** framework. Rather than a single independent module, CATF acts as a joint optimization strategy that guides both the front-end cost fusion and the back-end penalty adjustment. As illustrated in Figure 1, the fully pipelined architecture comprises three core stages seamlessly connected via on-chip line buffers:

- (1) **Texture-Guided Cost Computation & Fusion:** Computes initial matching costs for RGB and NIR using the Census transform, synchronously extracts texture confidence via SAD-Center, and adaptively fuses the dual-modal costs.
- (2) **Epipolar Redundancy Mechanism (ERM):** Activates a hardware-efficient multi-line parallel search guided by texture intensity, mitigating dynamic vertical residuals caused by mechanical or thermal drifts.
- (3) **Adaptive-Penalty Streamlined SGM:** Performs 4-path cost aggregation where the  $P_2$  penalty parameters are dynamically regulated by the shared CATF texture confidence, followed by standard WTA and filtering for final disparity map generation.

### 3.1 Texture-guided Cost Computation & Fusion

To counteract the significant non-linear radiometric differences between RGB and NIR modalities, we adopt the Census transform, which is inherently invariant to monotonic intensity variations. For a center pixel  $p$  and its neighborhood  $N(p)$ , the standard Census descriptor  $B(p)$  is generated by concatenating the boolean comparison results between the intensity  $I(p)$  and surrounding pixels. The raw matching cost  $C_{raw}^k(p, d)$  for modality  $k \in \{RGB, NIR\}$  at disparity  $d$  is then computed as the Hamming distance between left and right descriptors.

While Census handles global illumination variations, its signatures become unreliable in textureless or low-SNR regions (e.g., RGB shadows or NIR-absorbing fabrics). Under such conditions, fixed-weight fusion allows the degraded modality to corrupt the reliable matching cost. Thus, cost-domain adaptive fusion guided by local texture confidence is essential.

To evaluate the credibility of pixel  $p$  under modality  $k$  without extra BRAM overhead, we propose to directly reuse the existing Census support window  $N(p)$ . While traditional edge operators (e.g., Sobel or Laplacian) require complex multiplication logic to compute gradients, we employ the Sum of Absolute Differences to Center (SAD-Center) as the texture intensity metric  $E_{tex}^k(p)$ :

$$E_{tex}^k(p) = \sum_{q \in N(p)} |I^k(q) - I^k(p)| \quad (1)$$

Statistically,  $E_{tex}$  represents the total absolute deviation within the local window. By relying exclusively on simple subtraction and accumulation, this lightweight metric effectively assigns high confidence to textured regions (higher  $E_{tex}$ ) while suppressing textureless areas (lower  $E_{tex}$ ). Consequently, it achieves robust texture evaluation while adding negligible combinational logic overhead to the existing Census datapath.

To translate  $E_{tex}$  into a normalized matching confidence  $\alpha^k \in [0, 1]$  with soft-thresholding characteristics, we employ a Sigmoid mapping kernel:

$$\alpha^k(p) = \frac{1}{1 + e^{-\lambda(E_{tex}^k(p) - \mu)}} \quad (2)$$

where  $\lambda$  controls the sensitivity to texture variations, and  $\mu$  serves as the baseline decision threshold. For hardware efficiency, this non-linear function is implemented via a pre-computed Look-Up Table (LUT).

Finally, the adaptive fused matching cost  $C_{fused}$  is calculated as the confidence-weighted sum:

$$C_{fused}(p, d) = \frac{\alpha^{rgb} \cdot C_{raw}^{rgb}(p, d) + \alpha^{nir} \cdot C_{raw}^{nir}(p, d)}{\alpha^{rgb} + \alpha^{nir} + \epsilon} \quad (3)$$

This dynamic fusion mechanism executes pixel-level modal selection: in RGB shadowed regions ( $\alpha^{rgb} \approx 0$ ), the fused cost is dominated by the NIR modality, whereas in regions with complementary textures, it performs a weighted average to maximize the SNR.

### 3.2 Texture-Aware Epipolar Redundancy Mechanism (ERM)

In standard stereo systems, dynamic physical disturbances (e.g., thermal drift, mechanical vibration, and minor roll errors) inevitably

induce sub-pixel to 1-pixel vertical disparity residuals over time [9]. These residuals violate the strict 1D scanline epipolar constraint. Correcting these shifts via traditional image-domain dynamic remapping requires prohibitive DDR bandwidth. Instead, we propose an Epipolar Redundancy Mechanism (ERM) that compensates for these residuals directly in the cost domain, adopting a strategy of “tolerance over rectification.”

Building upon the static expansion concept in [19], our ERM relaxes the strict scanline constraint by expanding the cost search space vertically by  $\pm 1$  pixel. To implement this multi-line parallel search with minimal hardware footprint, we buffer the compressed Census bit-strings using two line buffers. By buffering feature descriptors instead of raw pixels or full cost volumes, the system achieves a 3-pixel vertical capture range with low BRAM overhead. Essentially acting as a  $3 \times 1$  morphological erosion operation within the cost domain, this mechanism effectively absorbs vertical jitters within a  $\pm 1$  pixel range.

However, blindly expanding the search window across the entire image exacerbates the risk of mismatches in textureless regions. To achieve a dynamic balance between geometric robustness and matching reliability, the ERM is designed as a hardware-selectable, texture-aware gating mechanism. Regulated by a global enable signal ( $En_{mode}$ ) and the locally extracted SAD-Center texture intensity ( $E_{tex}$  from Eq. 1), the ERM leverages the spatial rigidity between the coaxial modalities, using the NIR channel to steer the optimal epipolar offset  $k^*$ .

$$k^* = \arg \min_{k \in \{-1, 0, 1\}} C_{NIR}(x, y + k, d) \quad (4)$$

The final matching cost  $C_{final}$  is dynamically computed as:

$$C_{final}(x, y, d) = \begin{cases} C_{fused}(x, y + k^*, d), & \text{if } En_{mode} \text{ and } E_{tex}(x, y) > T_{high} \\ C_{fused}(x, y, d), & \text{otherwise} \end{cases} \quad (5)$$

When the device operates under severe vibration or aging, the system activates the redundant search ( $En_{mode} = 1$ ) exclusively for unique high-frequency edge regions ( $E_{tex} > T_{high}$ ). For flat regions susceptible to ambiguity, it seamlessly reverts to the strict single-line constraint to suppress noise interference.

### 3.3 Adaptive-Penalty Streamlined SGM

Standard SGM approximates 2D global smoothness by aggregating costs along multiple 1D paths, utilizing a penalty  $P_2$  for depth discontinuities. However, a globally fixed  $P_2$  faces an inherent dilemma: textureless regions require strong penalization to suppress noise, whereas object boundaries demand weak penalization to preserve sharp contours.

To achieve hardware-efficient adaptive penalization, we directly reuse the modality confidence  $\alpha^k$  (Eq. 2) and texture intensity  $E_{tex}^k$  (Eq. 1) extracted by the CATF front-end. To prevent high noise in degraded or overexposed regions from being misclassified as structural edges, we gate the texture metric using the confidence

score and extract a robust cross-modal gradient,  $E_{robust}$ :

$$E_{robust}(x, y) = \max_{k \in \{RGB, NIR\}} \begin{cases} E_{tex}^k(x, y), & \text{if } \alpha^k(x, y) > T_{conf} \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

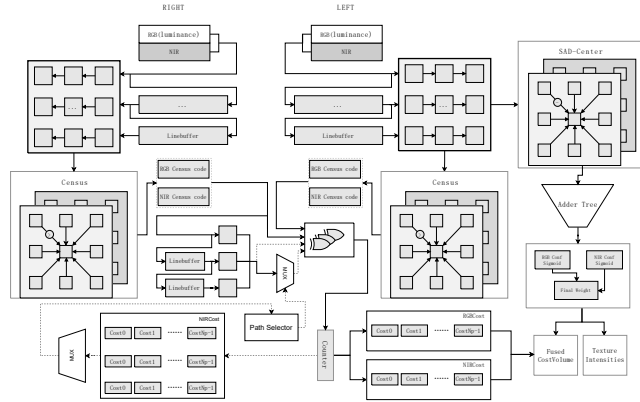
where  $T_{conf}$  is the validity threshold. This logic ensures that as long as either modality detects a reliable contour, the structural information is preserved. Based on  $E_{robust}$ , the penalty  $P_2$  is dynamically mapped to three discrete levels via hardware-friendly comparators:

$$P_2(x, y) = \begin{cases} P_{2,flat}, & \text{if } E_{robust}(x, y) < T_{low} \\ P_{2,edge}, & \text{if } E_{robust}(x, y) > T_{high} \\ P_{2,mid}, & \text{otherwise} \end{cases} \quad (7)$$

where  $P_{2,flat} > P_{2,mid} > P_{2,edge}$ . This parameter transition requires zero additional storage and is fully embedded within the pipeline.

Furthermore, traditional 8- or 16-path SGM requires backward aggregation (e.g., bottom-to-top), which explicitly violates the raster scan order and necessitates external DDR frame buffers. To strictly adhere to a streaming architecture, we prune the aggregation to four forward-facing paths ( $r \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$ ) that depend exclusively on previously buffered rows.

After aggregating the directional costs using the relaxed data term  $C_{final}$  (Eq. 5) and the adaptive  $P_2$ , the final disparity is determined via a standard Winner-Takes-All (WTA) strategy. Finally, a uniqueness check (Peak-to-Second-Peak Ratio  $< \tau_{uniq}$ ), a left-to-right background extension to fill holes and a  $3 \times 3$  median filter are applied within the data stream to eliminate repetitive ambiguities and salt-and-pepper noise, completing the fully DDR-less disparity computation pipeline.



**Figure 2: Overall RTL architecture of the hardware-optimized CATF front-end and epipolar routing.**

## 4 RTL Implementation and Timing Optimization

Mapping the proposed CATF framework and ERM onto a resource-constrained FPGA involves a strict trade-off between processing throughput and hardware utilization. To sustain a 1 pixel-per-clock (PPC) raster-scan throughput, the datapath leverages standard BRAM line buffers. This section details its RTL mapping, focusing

on shift-register management and critical-path retiming optimizations for real-time processing.

### 4.1 Hardware-Optimized CATF Front-End and Epipolar Routing

Acting as the input datapath, this module computes matching costs from the raw RGB-NIR streams. The 24-bit RGB input is first converted to 8-bit grayscale via fixed-point shifts:  $Y = (77R + 150G + 29B) \gg 8$ . The resulting grayscale data is then cycle-aligned with the NIR stream.

As shown in Fig. 2, dual Census cores are instantiated to process the heterogeneous streams in parallel. RGB luminance and NIR data are concatenated into a unified line buffer. A 24-way parallel comparator array generates the dual-modal 24-bit Census bitstrings within a single clock cycle. To compute the SAD-Center metric, the  $5 \times 5$  Census register window is directly reused via a pipelined adder tree.

Directly implementing the non-linear Sigmoid function requires complex exponential and division operations, incurring significant hardware overhead and deep pipeline latency. Instead, it is mapped to an offline pre-computed 256-byte ROM. The 13-bit SAD is right-shifted by 2 bits to filter LSB noise and saturated to 8 bits, naturally aligning with the Sigmoid saturation curve. Subsequently, a dual-compression strategy is employed to generate the fusion weight  $w_{rgb}$ . Using the 4 MSBs of both confidences as address indices compresses the 2D fusion LUT from a full-precision  $256 \times 256$  table down to a  $16 \times 16$  LUTRAM. Quantizing the output weight  $w_{rgb}$  to  $[0, 32]$  replaces the subsequent normalization division with a 5-bit right shift.

Following temporal alignment, the fully unrolled Census module computes  $2N_p$  Hamming distances, where  $N_p$  is the disparity range, via parallel XOR arrays per cycle. The RGB and NIR costs are then fused using the quantized  $w_{rgb}$ . To handle invalid data during pipeline priming, boundary costs are clamped to a saturated maximum.

The ERM's  $\pm 1$  vertical search window is generated directly within the compressed Census domain (Fig. 2). The right-image feature stream cascades through two line buffers, providing parallel access to the  $m_0$  ( $y+1$ ),  $m_1$  ( $y$ ), and  $m_2$  ( $y-1$ ) rows.

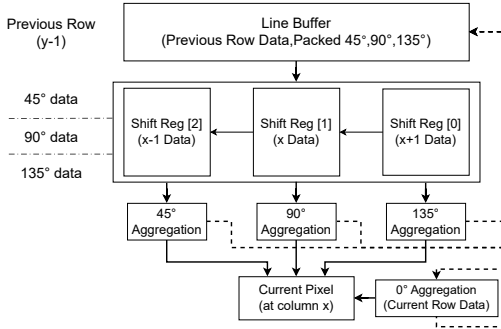
A multiplexer, governed by the global  $En_{mode}$  flag and the local  $E_{tex}^L > T_{high}$  condition, controls the texture-aware routing. When the global redundancy mode is disabled ( $En_{mode} = 0$ ), the system bypasses the cascaded buffer paths entirely, routing data through the standard datapath to align with the default pipeline latency. When the mode is globally enabled ( $En_{mode} = 1$ ), the feature stream is routed through the cascaded buffers. Within this active state, if the pixel exhibits high texture ( $E_{tex}^L > T_{high}$ ), the NIR modality first identifies the minimum-cost path among the  $m_0$ ,  $m_1$ , and  $m_2$  rows, which then steers a multiplexer to select the corresponding raw RGB Census code prior to its Hamming computation, dynamically compensates for vertical errors. Conversely, in textureless regions ( $E_{tex}^L \leq T_{high}$ ), the logic strictly outputs the matching cost of the center row ( $m_1$ ) to enforce epipolar constraints and prevent ambiguity. This logic enables robust fault tolerance with two line buffers.

## 4.2 Texture-Aware Streaming SGM Accelerator

To map the SGM algorithm onto a resource-constrained edge FPGA without external memory, a row-level pipelined 4-path aggregation engine is proposed.

The adaptive  $P_2$  penalty is generated via a cross-modal maximization logic. The 8-bit confidences of both channels are first compared against  $T_{conf}$  to gate invalid SAD values. Parallel comparators then select the maximum texture intensity across the dual modalities. A single-stage combinational logic maps this intensity to three discrete penalty levels ( $P_{2,edge}, P_{2,mid}, P_{2,flat}$ ), preventing over-smoothing in low-SNR regions while preserving sharp depth discontinuities.

The aggregation core utilizes a fully unrolled architecture, instantiating  $N_p$  Processing Elements (PEs) concurrently to compute all disparity hypotheses within a single clock cycle. Each PE implements Add-Compare-Select (ACS) logic for four forward paths ( $0^\circ, 45^\circ, 90^\circ, 135^\circ$ ). To bound BRAM utilization and prevent accumulator overflow, path costs are saturated to 11 bits.



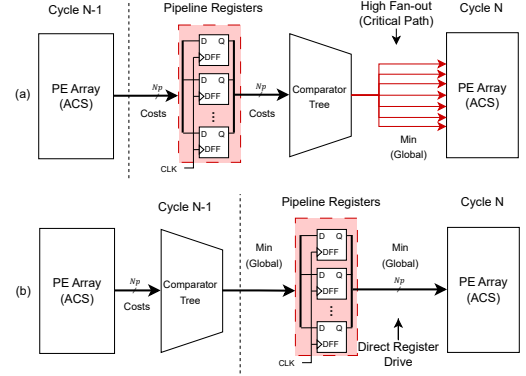
**Figure 3: Sliding-window shift-register architecture for strict inter-row context alignment.**

Managing parallel data dependencies for diagonal paths without stalling the 1 PPC pipeline requires precise spatial alignment. As illustrated in Fig. 3, dual-port BRAMs configured as line buffers store the previous row's path costs. A pre-fetching sliding-window mechanism is employed: the BRAM read address advances one cycle ahead of the current pixel. This data feeds a 3-tap cascaded shift register. By tapping different register stages concurrently, the cycle-aligned context costs for  $x + 1$  ( $135^\circ$  path),  $x$  ( $90^\circ$  path), and  $x - 1$  ( $45^\circ$  path) are extracted seamlessly to feed the subsequent ACS logic.

## 4.3 Critical Path Retiming and Post-Processing

While inter-row paths ( $45^\circ, 90^\circ, 135^\circ$ ) are fully pipelined, the  $0^\circ$  intra-row path introduces a severe recursive data dependency, restricting the maximum operating frequency ( $F_{max}$ ). Specifically, computing the current path cost  $L_r(p, d)$  strictly depends on the global minimum of the preceding pixel, expressed as:

$$L_r(p, d) = C_{final}(p, d) + \min \begin{cases} L_r(p - r, d) \\ L_r(p - r, d \pm 1) + P_1 \\ \min_k L_r(p - r, k) + P_2 \\ - \min_k L_r(p - r, k) \end{cases} \quad (8)$$



**Figure 4: Comparison of the intra-row cost aggregation architectures .(a)Standard  $0^\circ$  critical path with high comparator fan-out. (b) Retimed  $0^\circ$  critical path with look-ahead logic. The retiming strategy drastically reduces the combinational path delay.**

As depicted in Fig. 4(a), the standard 1 PPC implementation requires a deep  $N_p$ -to-1 comparator tree to calculate  $\min_k L_r(p - 1, k)$ . This result then feeds all  $N_p$  PEs within the same clock cycle to execute the ACS and normalization operations. This deep combinational logic chain generates extremely high fan-out and severe setup-time violations.

To resolve this, we employ a retiming strategy by shifting the pipeline boundary (Fig. 4(b)). Instead of computing the global minimum at the beginning of the current ACS cycle, this calculation is shifted to the end of the previous cycle. During the same clock cycle that the  $N_p$  aggregation costs are computed, the comparator tree calculates the global minimum of the current pixel. This minimum is latched into a pipeline register concurrently with the aggregated costs. In the subsequent clock cycle, the ACS normalization logic reads this value directly from the register. This structural shift decouples the comparator tree from the PE input stage, drops its combinational output load to a single register, and eliminates the massive routing delay caused by high fan-out. Coupled with register duplication, this RTL optimization significantly boosts  $F_{max}$  while maintaining strict algorithmic equivalence.

Finally, a tightly coupled post-processing pipeline refines the disparity map. A 6-stage pipelined comparator tree executes the Winner-Takes-All (WTA) decision. Simultaneously, a uniqueness check ( $C_{sec} \geq C_{min} + (C_{min} \gg N)$ ) mitigates repetitive ambiguities using hardware right-shifts. For invalidated pixels, a Left-to-Right Background Extension strategy propagates the last reliable background disparity via a holding register. This strictly forward-feeding approach maintains the continuous dataflow and effectively prevents foreground "fattening". A parallel  $3 \times 3$  sorting network subsequently applies median filtering to eradicate salt-and-pepper noise, outputting the final dense depth map.

## 5 Experimental Results and Analysis

The proposed architecture is implemented in Verilog and synthesized on a Xilinx Zynq-7020 SoC using Vivado 2021.2, with a maximum disparity range of 64 pixels. Evaluation is conducted on the

Pixel-aligned RGB-NIR Stereo Dataset (CVPR 2025). After rigorously filtering out invalid LiDAR reference frames (due to sensor over-saturation or severe misalignment), the final test set comprises 669 valid frames ( $720 \times 540$ ) across eight sequences, covering low-light, specular, and standard outdoor scenarios.

System performance is evaluated across two dimensions: (1) **Geometric Quality**: Quantified by Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Bad Pixel Rate (BPR,  $\delta > 5$ ). Given inherent real-world calibration residuals, our evaluation emphasizes relative performance gains against baselines rather than absolute magnitudes. (2) **Hardware Efficiency**: Measured by resource utilization (LUTs, FFs, BRAMs), power consumption, and throughput (FPS) to verify suitability for edge deployment.

## 5.1 Algorithm Performance Evaluation

To validate the algorithmic efficacy of the proposed architecture prior to hardware deployment, we evaluate the system using a custom bit-accurate Python model that strictly emulates the fixed-point RTL behavior. We compare our CATF system against two baselines: (1) the standard visible-only OpenCV SGBM, and (2) a multimodal OpenCV SGBM utilizing a naive pixel-level average fusion input. All output disparity maps are uniformly cropped by 64 pixels from the left edge to exclude the inherent invalid matching region.

**Qualitative Analysis.** As visually demonstrated in Fig. 5, the proposed CATF framework exhibits consistent robustness across three challenging unstructured environments, effectively handling distinct single- or dual-modality failure conditions:

**RGB Failure (Row 1, Extreme Low-light):** Due to the severe SNR degradation of the visible sensor, the baseline RGB SGBM suffers from significant accuracy loss, producing severe streaking artifacts and failing to reconstruct large-scale structures (e.g., trees). While naive pixel-fusion introduces the NIR band to recover brightness, it incorporates severe specular reflection noise from the smooth marble ground into the cost volume. In contrast, our CATF module effectively utilizes the NIR modality to reconstruct background vegetation contours, while the adaptive  $P_2$  penalty automatically heavily penalizes the textureless marble surface, suppressing the reflection-induced mismatch artifacts.

**NIR Failure (Row 2, Artificial Light Interference):** In this scenario, the highly reflective floor inherently violates the Lambertian assumption and causes severe overexposure and false textures in the NIR spectrum. Consequently, the naive pixel-fusion strategy is significantly compromised by the NIR interference, distorting the originally reliable RGB cost minima. Our algorithm detects this cross-modal gradient discrepancy. By assigning a near-zero texture confidence to the overexposed NIR region, the CATF adaptively suppresses the NIR weight, allowing the disparity computation to be robustly dominated by the unaffected RGB stream.

**Dual-Modality Failure (Row 3, Specular Glass Wall):** Daytime glass walls represent a highly ill-posed condition, where high transmittance eliminates structural texture, and view-dependent surface reflections disrupt photometric consistency across both spectra. Both baseline methods experience severe matching failures, exhibiting significant depth discontinuities. Facing this dual-modality degradation, our system extracts a near-zero  $E_{robust}$  metric. This

triggers the  $P_{2,flat}$  strong smoothing constraint, actively propagating correct depth hypotheses from the surrounding high-texture window frames into the textureless glass region. Concurrently, the penalty dynamically relaxes at the structural boundaries on the right, strictly preserving sharp depth edges.

In summary, rather than blindly fusing multimodal pixels, the proposed CATF framework achieves dynamic modality blending and structural regularization, elevating the reliability of edge vision systems in diverse lighting conditions.

**Quantitative Analysis.** Table 1 summarizes the quantitative evaluation under  $720 \times 540$  resolution and 64-pixel disparity search constraint. On average, the proposed CATF achieves superior overall geometric recovery accuracy.

**Performance in Well-lit Environments:** In well-lit scenarios, our method reduces the Mean Absolute Error (MAE) to 5.67 px, marking a 16.0% and 10.8% improvement over the RGB-only and pixel-fusion baselines, respectively. This confirms that adaptive fusion within the cost domain strictly preserves high-frequency structural details, effectively circumventing the texture blurring intrinsically introduced by naive pixel-level blending.

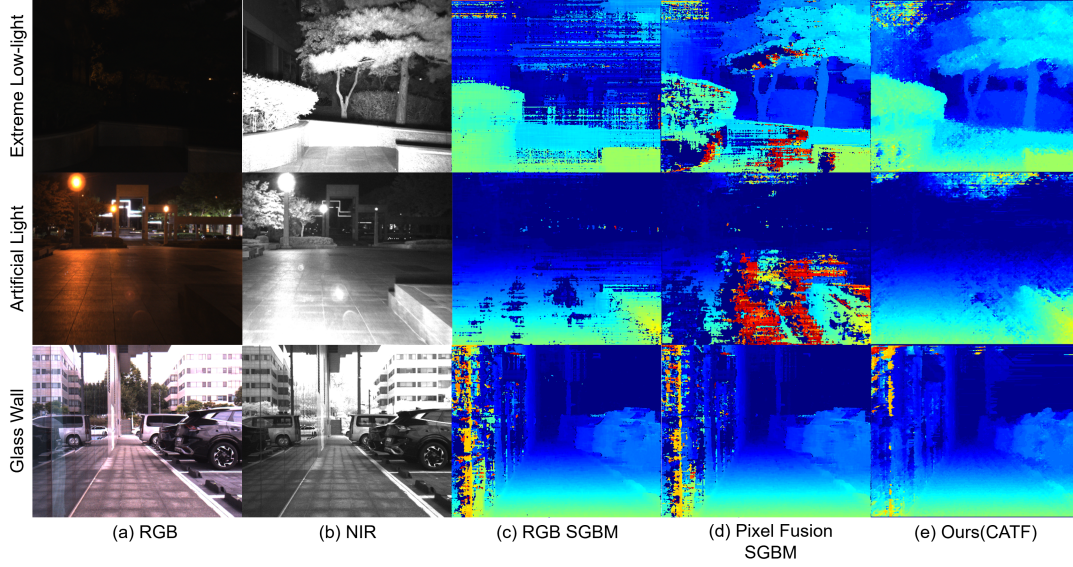
**Robustness in Low-light Environments:** The architectural advantages become paramount in degraded conditions. While the RGB-only MAE diverges severely (7.38 px) due to ambient illumination drops, our CATF maintains an MAE of 5.91 px—nearly mirroring its reference well-lit performance. Notably, although the pixel-fusion baseline achieves a marginally lower BPR (32.00%) via simplistic noise averaging, its overall geometric deviation (MAE 6.68 px) remains significantly worse. This discrepancy reveals that blind pixel-level averaging irreversibly blurs object contours. Conversely, the CATF strategy exhibits a decisive advantage in preserving edge sharpness and global structural consistency.

**Table 1: Quantitative accuracy comparison across diverse lighting scenarios.**

| Scenario | Algorithm Method        | MAE (px)    | RMSE (px)   | BPR (>5px)    |
|----------|-------------------------|-------------|-------------|---------------|
| Well-lit | Baseline (RGB-only)     | 6.75        | 10.24       | 28.65%        |
|          | Baseline (Pixel-Fusion) | 6.36        | 9.78        | 26.88%        |
|          | <b>Ours (CATF)</b>      | <b>5.67</b> | <b>8.86</b> | <b>25.19%</b> |
| Dark     | Baseline (RGB-only)     | 7.38        | 10.63       | 37.58%        |
|          | Baseline (Pixel-Fusion) | 6.68        | 9.73        | <b>32.00%</b> |
|          | <b>Ours (CATF)</b>      | <b>5.91</b> | <b>8.60</b> | 33.67%        |

**Hardware-Accuracy Trade-off Analysis.** To quantify the overhead and performance gains of the proposed modules, we designed targeted ablation experiments, evaluated on three specific subsets: (1) *Weak-texture* (5 sequences of smooth walls/low-light); (2) *Low-SNR* (well-lit sequences injected with  $\sigma = 30$  Gaussian noise and brightness attenuation); and (3) *Misalignment* (115 frames with manually injected 1-pixel right-eye vertical shifts mimicking thermal drift). In this analysis, the **Baseline** configuration refers to a standard fully-unrolled SGM datapath utilizing a fixed-weight pixel fusion ( $w = 0.5$ ) without the ERM. Results and corresponding physical hardware overheads are detailed in Table 2.

**Cost-Efficiency of CATF:** Synthesized on the Zynq-7020 SoC, the CATF datapath incurs a minimal logic overhead of approximately 600 LUTs ( $\sim 1.5\%$  of the total logic) and requires zero additional BRAM. This efficiency is achieved by mapping the compressed 2D



**Figure 5: Qualitative comparison of disparity maps in ill-posed outdoor scenarios. The columns evaluate baseline methods against our CATF, while the rows demonstrate system robustness under three distinct cross-modal failure modes (RGB, NIR, and dual-modality failure).**

fusion table to distributed RAM. Compared to the fixed-weight pixel fusion ( $w = 0.5$ ), CATF yields a relative BPR reduction of 17.1% (from 31.49% to 26.09%) in challenging weak-texture regions. This confirms the necessity of texture-guided adaptive cost aggregation in correcting disparity search directions. Under severe Low-SNR conditions, CATF outperforms the baseline by 18.35% in BPR. This demonstrates an efficient balance between hardware resources and accuracy: adaptively fusing modalities in the cost domain preserves the integrity of NIR features, preventing irreversible signal corruption with negligible logic overhead.

*Cost-Efficiency of ERM:* Activating the runtime-configurable ERM requires instantiating parallel Hamming distance pipelines across the  $\pm 1$  vertical search space for all 64 disparity hypotheses, incurring an additional logic overhead by roughly 8,000 LUTs alongside 4.5 BRAM tiles. Evaluated on the misalignment subset, activating the ERM successfully recovers the matching accuracy (MAE drops from 6.96 to 6.10 px), yielding a 21.1% relative reduction in BPR (from 28.51% to 22.49%). Absorbing physical geometric drifts directly within the feature-domain stream using pure combinational logic bypasses the need for computationally prohibitive and DSP intensive stereo rectification engines.

**Table 2: Hardware-accuracy trade-off analysis across extreme test subsets.**

| Method   | Overhead<br>LUTs/BRAMs | Weak-Tex.<br>MAE/BPR | Low-SNR<br>MAE/BPR   | Misalign.<br>MAE/BPR |
|----------|------------------------|----------------------|----------------------|----------------------|
| Baseline | —                      | 6.17 / 31.49%        | 8.80 / 52.09%        | —                    |
| + CATF   | ~600 / 0               | <b>5.51 / 26.09%</b> | <b>6.67 / 33.74%</b> | 6.96 / 28.51%        |
| + ERM    | ~8,600 / 4.5           | —                    | —                    | <b>6.10 / 22.49%</b> |

## 5.2 Hardware Implementation and Comparison

The proposed CATF architecture (excluding the optional ERM) was implemented and verified on the Xilinx Zynq-7020 SoC. Table 3 presents the hardware utilization and performance metrics alongside existing FPGA-based stereo matching accelerators.

**Table 3: Hardware Utilization Comparison with Existing FPGA Implementations**

| Work            | [1]      | [4]       | [25]       | Ours      |
|-----------------|----------|-----------|------------|-----------|
| Modality        | RGB      | RGB       | RGB        | RGB-NIR   |
| Resolution      | 640×480  | 640×480   | 1920×1080  | 720×540   |
| Disparity Range | 64       | 128       | 256        | 64        |
| Platform        | Virtex-5 | Spartan-7 | Stratix-IV | Zynq-7020 |
| LUTs            | 68,427   | 29,797    | 298,397    | 35,868    |
| FFs             | —        | 31,327    | 94,238     | 22,282    |
| Memory (bits)   | 3.22M    | 622K      | 2.18M      | 2.43M     |
| Power (W)       | —        | 0.517     | —          | 0.825     |
| FPS             | 66       | 355       | 50.15      | 128       |
| Frequency (MHz) | 60       | 109       | 104        | 50        |

*Architecture and Efficiency Analysis:* As shown in Table 3, the DDR-based buffering in [1] supports multi-directional aggregation but incurs prohibitive external memory and power. In contrast, targeting ultra-low power, [4] reduces the aggregation paths to only two, which significantly degrades depth precision at object boundaries. While [25] utilizes min-pooling to compress memory, the resulting complex datapaths inflate LUT consumption by over 1.5×, exceeding the capacity of most edge FPGAs.

Our system eliminates external memory reliance through a deeply optimized streaming line-buffer architecture. By leveraging feature-domain cascaded buffering and a compact 4-path aggregation strategy, the fully unrolled, memory-intensive SGM is successfully mapped onto the constrained Zynq-7020, consuming 35,868 LUTs and 2.43 Mb of BRAM (47% utilization). Notably, despite supporting

four full-aggregation paths and dual-modality processing, our architecture maintains a lower Flip-Flop (FF) footprint (22,282) compared to the two-path design in [4]. Operating at 1 pixel/clock, the design achieves a high throughput of 128 FPS at 50 MHz for 720×540 resolution. Beyond resource efficiency, the core advantage lies in its cross-modal adaptability: to our knowledge, this is the first FPGA-based accelerator to realize pixel-aligned RGB-NIR stereo matching. The texture-aware gating and adaptive penalty modulation enable all-day perception with negligible logic overhead, demonstrating that joint optimization of hardware and algorithms can achieve high robustness on edge devices while maintaining low latency.

### 5.3 Summary

The evaluation results demonstrate that the proposed accelerator significantly enhances depth estimation robustness in degraded lighting environments via RGB-NIR complementarity. Concurrently, the hardware implementation establishes an optimized balance between logic area, memory footprint, and perceptual reliability, offering a competitive hardware paradigm for all-day edge vision in mobile robotics.

## 6 Conclusion

To address the challenges of complex illumination and strict resource constraints in edge robotics, this paper proposed a hardware-algorithm co-designed RGB-NIR stereo matching architecture for co-axial sensors. By leveraging the physical pixel-alignment characteristic of the sensors, we developed a DDR-less full-pipeline architecture that fundamentally breaks the memory wall bottleneck in heterogeneous data processing. Building upon this, a Cost-domain Adaptive Texture Fusion (CATF) algorithm was proposed, which deeply reuses the Census transform array to extract texture confidence with zero extra memory overhead and adaptively adjusts SGM penalty parameters. Furthermore, an Epipolar Redundancy Mechanism (ERM) was designed at the hardware level to tolerate dynamic vertical residuals up to one pixel, effectively mitigating errors caused by mechanical vibrations and thermal drifts. RTL synthesis evaluation demonstrates that the proposed system consumes only 35,868 LUTs while achieving high-throughput real-time processing of 720 × 540 video streams. Compared to the traditional OpenCV SGBM algorithm, our architecture reduces the MAE by up to 19.9% in all-day scenarios, achieving an exceptional balance between high-accuracy depth perception and extreme resource efficiency on strictly constrained edge computing platforms.

## Acknowledgments

This work was supported by the Tianjin Major Special Project for Artificial Intelligence Science and Technology under Grant No. 25ZXRGX00100.

## References

- [1] Christian Banz, Sebastian Hesselbarth, Holger Flatt, et al. 2010. Real-time stereo vision system using semi-global matching disparity estimation: Architecture and FPGA-implementation. In *2010 International Conference on Embedded Computer Systems: Architectures, Modeling and Simulation*. 93–101. doi:10.1109/ICSAMOS.2010.5642077
- [2] Tzu-Yi Chuang, Hao-Wei Ting, and Jen-Jer Jaw. 2018. Dense Stereo Matching With Edge-Constrained Penalty Tuning. *IEEE Geoscience and Remote Sensing Letters* 15, 5 (2018), 664–668. doi:10.1109/LGRS.2018.2805916
- [3] Amaury Dame and Eric Marchand. 2012. Second-Order Optimization of Mutual Information for Real-Time Image Registration. *IEEE Transactions on Image Processing* 21, 9 (2012), 4190–4203. doi:10.1109/TIP.2012.2199124
- [4] Pingcheng Dong, Zhuoyu Chen, Zhuoao Li, et al. 2022. A 4.29nJ/pixel Stereo Depth Coprocessor With Pixel Level Pipeline and Region Optimized Semi-Global Matching for IoT Application. *IEEE Transactions on Circuits and Systems I: Regular Papers* 69, 1 (2022), 334–346. doi:10.1109/TCSI.2021.3100071
- [5] Wade S. Fife and James K. Archibald. 2013. Improved Census Transforms for Resource-Optimized Stereo Vision. *IEEE Transactions on Circuits and Systems for Video Technology* 23, 1 (2013), 60–73. doi:10.1109/TCSVT.2012.2203197
- [6] Maria Herrera-Arellano, Hayde Peregrina-Barreto, and Iván Terol-Villalobos. 2021. Visible-NIR Image Fusion Based on Top-Hat Transform. *IEEE Transactions on Image Processing* 30 (2021), 4962–4972. doi:10.1109/TIP.2021.3077310
- [7] Heiko Hirschmüller. 2008. Stereo Processing by Semiglobal Matching and Mutual Information. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 30, 2 (2008), 328–341. doi:10.1109/TPAMI.2007.1166
- [8] Han Hu, Yuri Rzhanov, Philip J. Hatcher, et al. 2015. Binary adaptive semi-global matching based on image edges. In *Seventh International Conference on Digital Image Processing (ICDIP 2015)*, Charles M. Falco and Xudong Jiang (Eds.), Vol. 9631. International Society for Optics and Photonics, SPIE, 96311D. doi:10.1117/12.2196960
- [9] Larisa Ivascu, Vlad-Florin Vinatu, and Mihail Gaianu. 2025. An Assessment of the Stereo and Near-Infrared Camera Calibration Technique Using a Novel Real-Time Approach in the Context of Resource Efficiency. *Processes* 13, 4 (2025). doi:10.3390/pr13041198
- [10] Xinyu Jia, Chuang Zhu, Minzhen Li, et al. 2021. LLVIP: A Visible-Infrared Paired Dataset for Low-Light Vision. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*. 3496–3504.
- [11] Seunghun Jin, Junguk Cho, Xuan Dai Pham, et al. 2010. FPGA Design and Implementation of a Real-Time Stereo Vision System. *IEEE Transactions on Circuits and Systems for Video Technology* 20, 1 (2010), 15–26. doi:10.1109/TCSVT.2009.2026831
- [12] Jinnyeong Kim and Seung-Hwan Baek. 2025. Pixel-aligned RGB-NIR Stereo Imaging and Dataset for Robot Vision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 11482–11492.
- [13] Jiayuan Li, Qingwu Hu, and Mingyao Ai. 2020. RIFT: Multi-Modal Image Matching Based on Radiation-Variation Insensitive Feature Transform. *IEEE Transactions on Image Processing* 29 (2020), 3296–3310. doi:10.1109/TIP.2019.2959244
- [14] Edgars Lielamurs, Andrejs Cvetkovs, Rihards Novickis, et al. 2023. Infrared Image Pre-Processing and IR/RGB Registration with FPGA Implementation. *Electronics* 12, 4 (2023). doi:10.3390/electronics12040882
- [15] Lahav Lipson, Zachary Teed, and Jia Deng. 2021. RAFT-Stereo: Multilevel Recurrent Field Transforms for Stereo Matching. In *2021 International Conference on 3D Vision (3DV)*. 218–227. doi:10.1109/3DV53792.2021.00032
- [16] Jialong Liu, Xichuan Zhou, Zhenlong Wan, et al. 2023. Multi-Scale FPGA-Based Infrared Image Enhancement by Using RGF and CLAHE. *Sensors* 23, 19 (2023). doi:10.3390/s23198101
- [17] Zhimin Lu, Jue Wang, Zhiwei Li, et al. 2022. A Resource-Efficient Pipelined Architecture for Real-Time Semi-Global Stereo Matching. *IEEE Transactions on Circuits and Systems for Video Technology* 32, 2 (2022), 660–673. doi:10.1109/TCSVT.2021.3061704
- [18] Jiayi Ma, Wei Yu, Pengwei Liang, et al. 2019. FusionGAN: A generative adversarial network for infrared and visible image fusion. *Information Fusion* 48 (2019), 11–26. doi:10.1016/j.inffus.2018.09.004
- [19] Phuc Hong Nguyen and Chang Wook Ahn. 2019. Stereo Matching Methods for Imperfectly Rectified Stereo Images. *Symmetry* 11, 4 (2019). doi:10.3390/sym11040570
- [20] Alex Noel Joseph Raj, M. Murugappan, and V. Arunachalam. 2021. Fusion of near-infrared and RGB images on a FPGA using high level synthesis tool. *Journal of Engineering Research* 9, 3, Part A (2021), 104–114. doi:10.36909/jer.v9i3A.9167
- [21] Ukcheol Shin, Jinsun Park, and In So Kweon. 2023. Deep Depth Estimation From Thermal Image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 1043–1053.
- [22] Xianqi Wang, Gangwei Xu, Hao Jia, et al. 2024. Selective-Stereo: Adaptive Frequency Information Selection for Stereo Matching. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 19701–19710.
- [23] Yina Wang, Mengjiao Gu, Yufeng Zhu, et al. 2022. Improvement of AD-Census Algorithm Based on Stereo Vision. *Sensors* 22, 18 (2022). doi:10.3390/s22186933
- [24] Zhenhao Yang, Yong Liang, Daoqian Lin, et al. 2024. Real-Time Stereo Vision Hardware Accelerator: Fusion of SAD and Adaptive Census Algorithm. *IEEE Access* 12 (2024), 154975–154989. doi:10.1109/ACCESS.2024.3479230
- [25] Wenye Zhang, Pingcheng Dong, Lei Chen, et al. 2025. Min-Pooling Cost Aggregation for Semi-Global Matching of Stereo Vision Processor. *IEEE Transactions on Circuits and Systems II: Express Briefs* 72, 1 (2025), 258–262. doi:10.1109/TCSII.2024.3463200